

PERSONAL PORTFOLIO

Dashboards and visualizations created using Python or MS power BI, IBM Analytics, Google Analytics

Darshit Chauhan

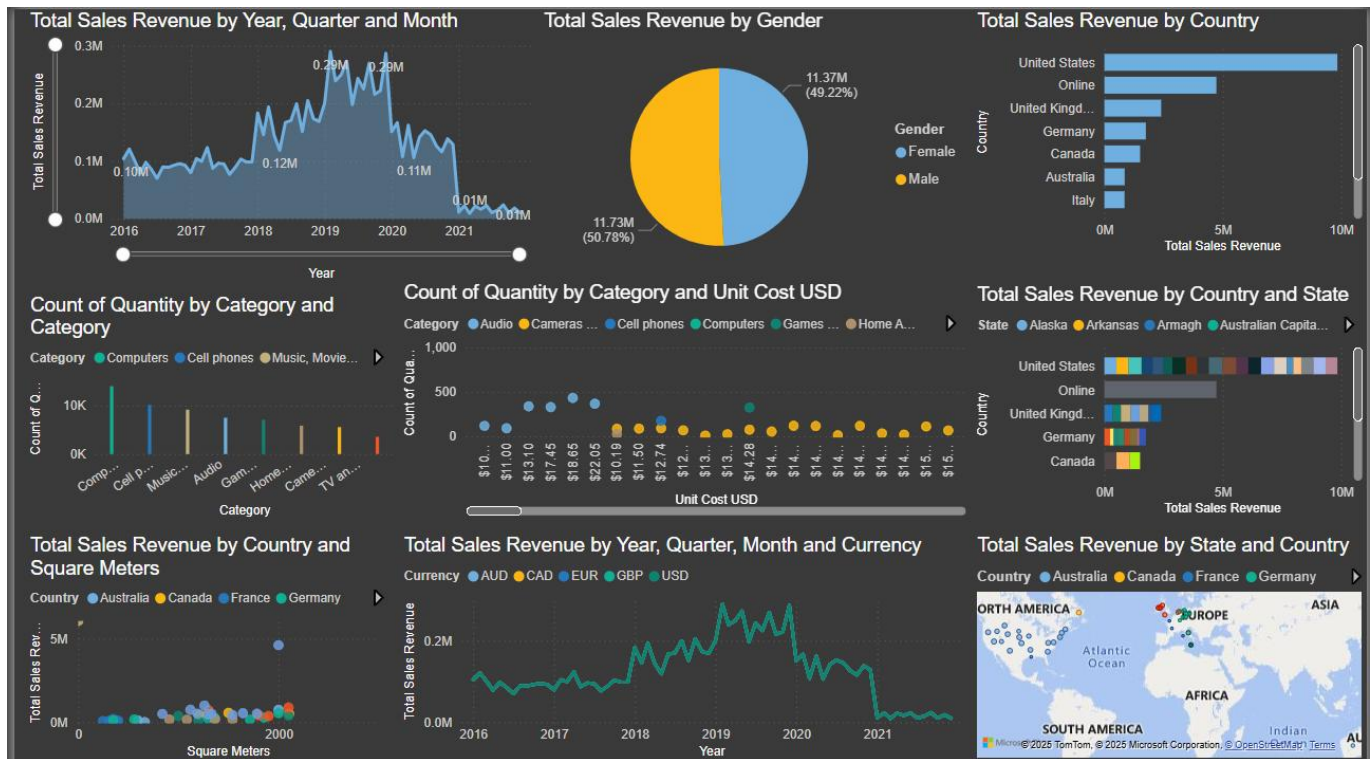
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1. Global Electronics retailer: (Multiple Table Dashboard using joins and DAX script in MS Power BI)

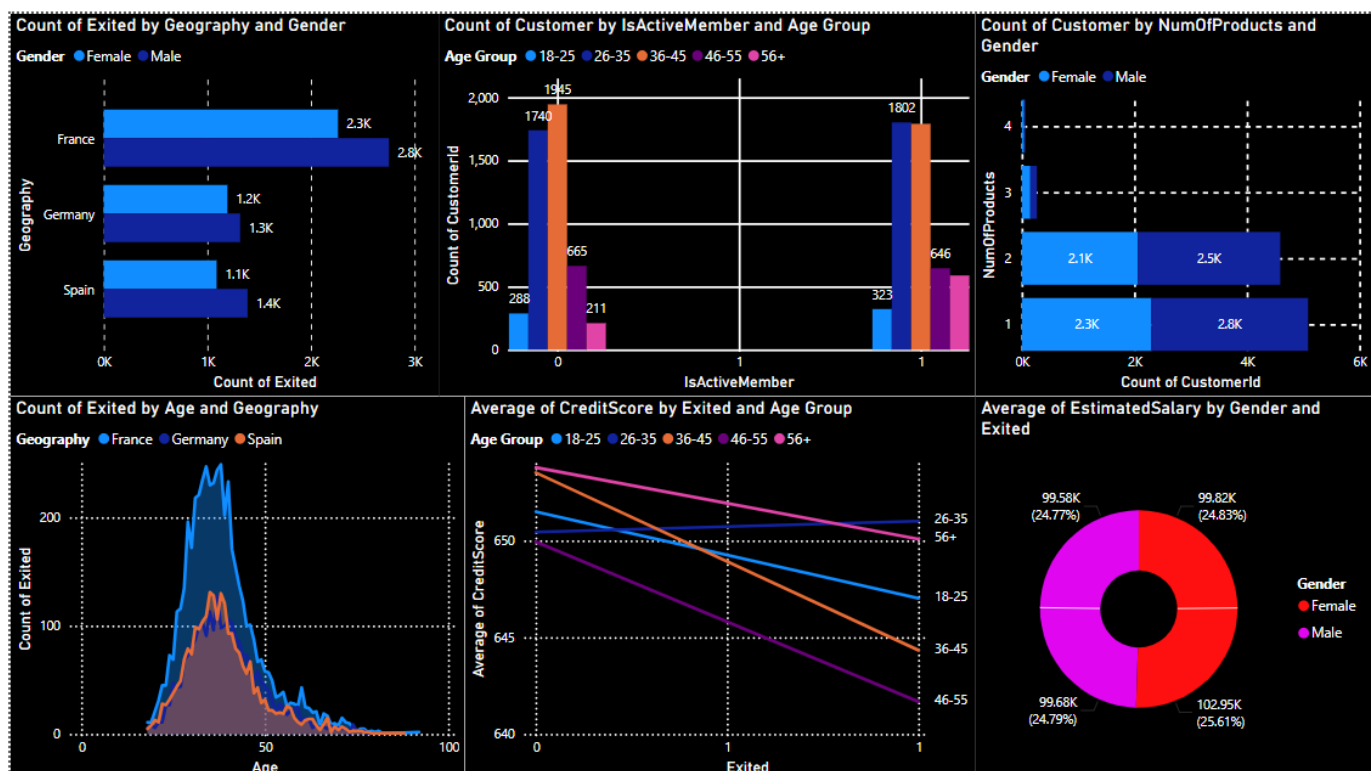
Dataset Source:

https://mavenanalytics.io/dataplayground?dataStructure=Multiple%20tables&order=date_added%2Cdesc&page=2&pageSize=5



2. Europe Bank Customer Churn data:

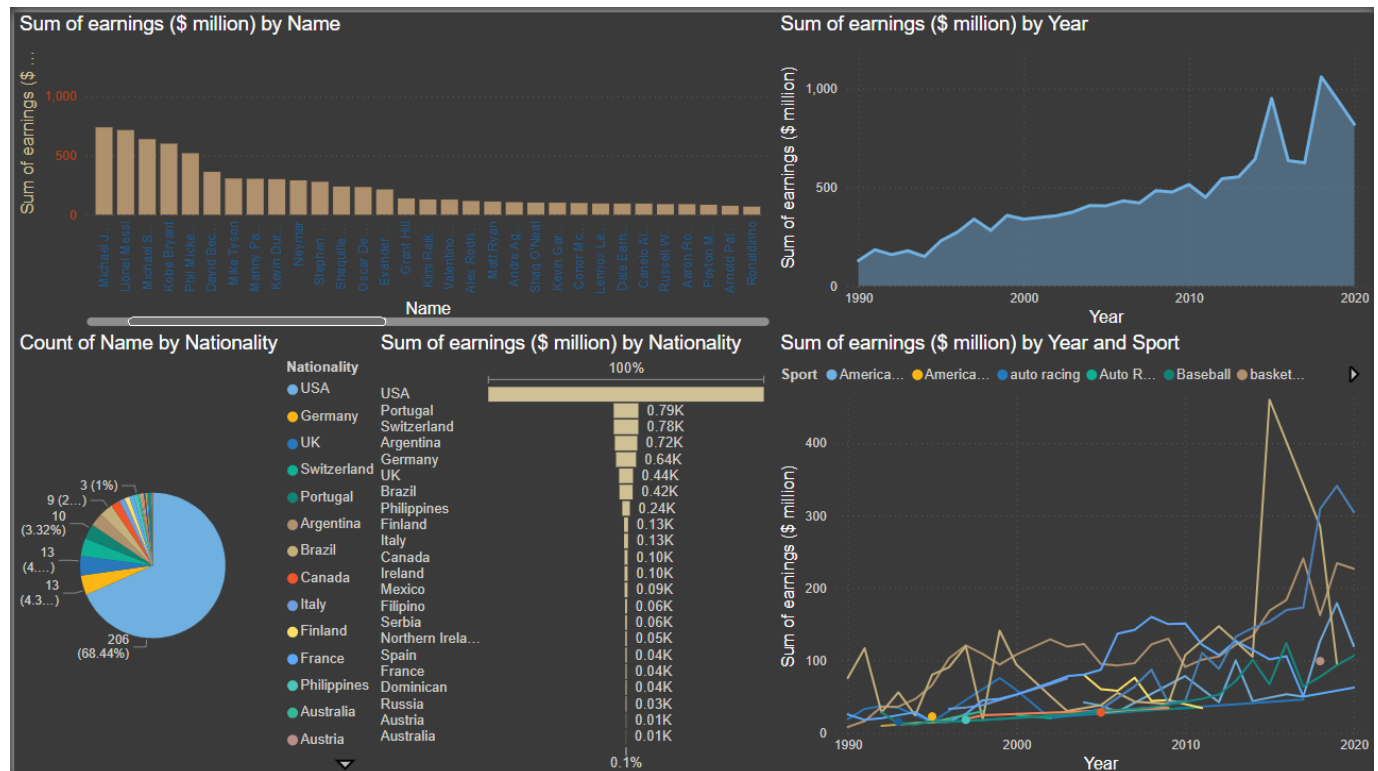
Dataset Source: <https://mavenanalytics.io/data-playground?page=2&pageSize=5>



3. Forbes Highest Paid Athletes (1990-2020).

Dataset source: <https://www.kaggle.com/datasets/parulpandey/forbes-highest-paid-athletes-19902019>

Tool used: MS Excel, MS Power BI



4. E-Commerce Data:

Dataset Source: <https://www.kaggle.com/datasets/shrishtimanja/ecommerce-dataset-for-data-analysis>

Tools Used: MS Power BI



5. Stock Market simulation (2022-2024)

Dataset source: <https://www.kaggle.com/datasets/samayashar/stock-market-simulation-dataset>

Tool used: Python (pandas, matplotlib, seaborn, plotly)

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import matplotlib.ticker as ticker

# Load the dataset
df = pd.read_excel("synthetic_stock_data.xlsx")

# ----- 1. Trend Distribution (Improved) -----
plt.figure(figsize=(6, 4))
ax = sns.countplot(data=df, x='Trend', palette='Set2')

# Increase the number of ticks for accuracy
ax.yaxis.set_major_locator(ticker.MultipleLocator(50))
plt.title("Market Trend Distribution")
plt.xlabel("Trend")
plt.ylabel("Count")
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()

# ----- 2. Sector-wise Metrics -----
sector_summary = df.groupby('Sector').agg({
    'Close': 'mean',
    'PE_Ratio': 'mean',
    'Dividend_Yield': 'mean'
}).round(2).reset_index()

# Close Price
fig1 = px.bar(sector_summary, x='Sector', y='Close', title='Average Close Price by Sector',
              color='Sector', text='Close')
fig1.update_layout(yaxis=dict(tickmode="linear", dtick=5)) # Increase y-axis tick intervals
fig1.show()

# PE Ratio
fig2 = px.bar(sector_summary, x='Sector', y='PE_Ratio', title='Average PE Ratio by Sector',
              color='Sector', text='PE_Ratio')
fig2.update_layout(yaxis=dict(tickmode="linear", dtick=2)) # Increase y-axis tick intervals
fig2.show()

# Dividend Yield
fig3 = px.bar(sector_summary, x='Sector', y='Dividend_Yield', title='Average Dividend Yield by Sector',
              color='Sector', text='Dividend_Yield')
fig3.update_layout(yaxis=dict(tickmode="linear", dtick=0.5)) # Increase y-axis tick intervals
fig3.show()

# ----- 3. Sentiment Score by Sector -----
sentiment_summary = df.groupby('Sector')['Sentiment_Score'].mean().reset_index().sort_values(by='Sentiment_Score', ascending=False)

plt.figure(figsize=(8, 4))
ax = sns.barplot(data=sentiment_summary, x='Sentiment_Score', y='Sector', palette='coolwarm')

# Increase the number of ticks for better accuracy
ax.xaxis.set_major_locator(ticker.MultipleLocator(0.1))
plt.title("Average Sentiment Score by Sector")
plt.xlabel("Avg Sentiment Score")
plt.ylabel("Sector")
plt.grid(axis='x', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()

# ----- 4. Sentiment vs Close Price -----
plt.figure(figsize=(6, 4))
ax = sns.scatterplot(data=df, x='Sentiment_Score', y='Close', hue='Sector', alpha=0.6)

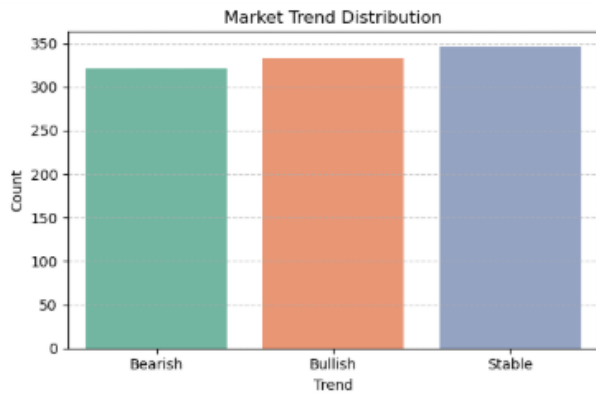
# More ticks for better accuracy
ax.xaxis.set_major_locator(ticker.MultipleLocator(0.2))
ax.yaxis.set_major_locator(ticker.MultipleLocator(5))
plt.title("Sentiment Score vs Close Price")
plt.xlabel("Sentiment Score")
plt.ylabel("Close Price")
plt.grid(True, linestyle='--', alpha=0.5)
plt.tight_layout()
plt.show()

# ----- 5. Time Series Close Price (Sample Company) -----
company_name = df['Company'].unique()[0]
company_df = df[df['Company'] == company_name]

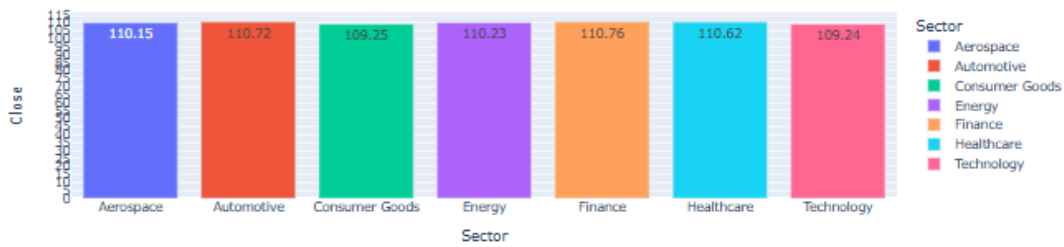
plt.figure(figsize=(10, 5))
ax = plt.plot(company_df['Date'], company_df['Close'], marker='o')

# Increase x-axis intervals for better readability
plt.xticks(rotation=45)
plt.locator_params(axis='x', nbins=10) # Increase the number of x-axis ticks

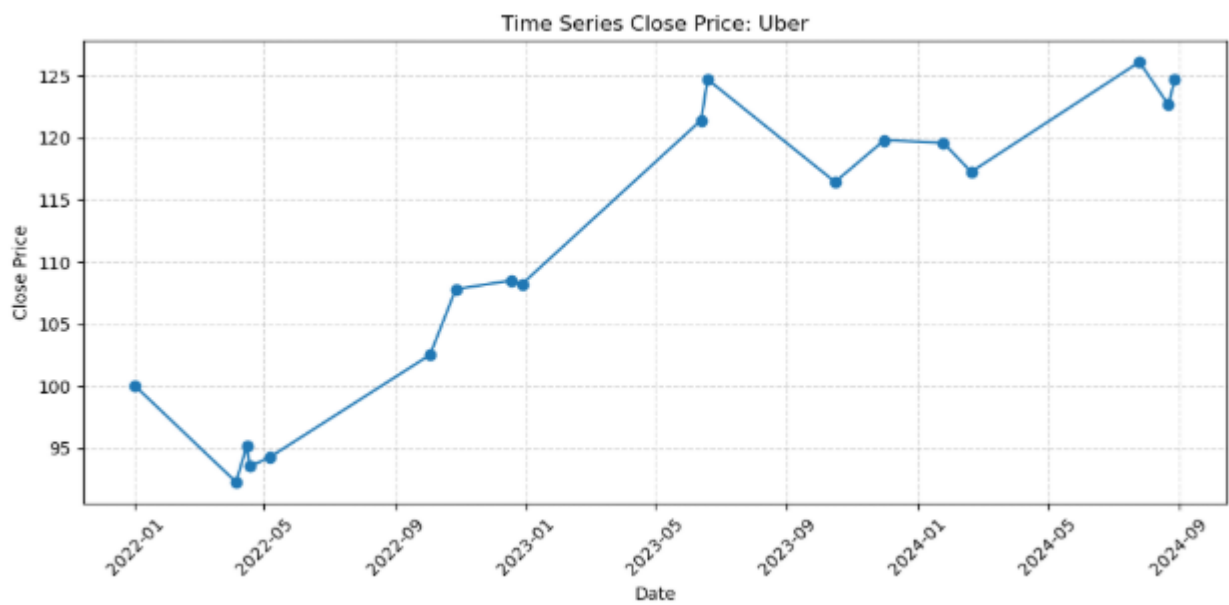
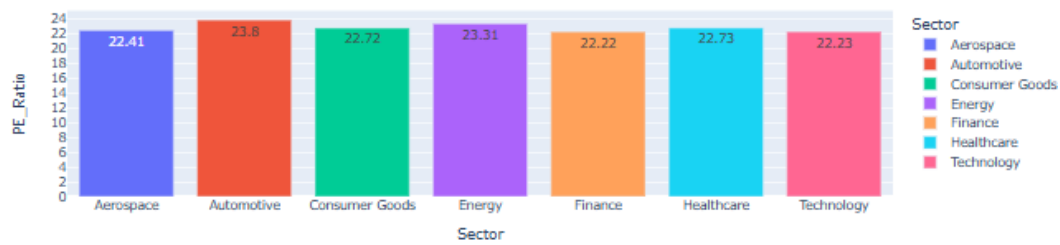
plt.title(f"Time Series Close Price: {company_name}")
plt.xlabel("Date")
plt.ylabel("Close Price")
plt.grid(True, linestyle='--', alpha=0.5)
plt.tight_layout()
plt.show()
```



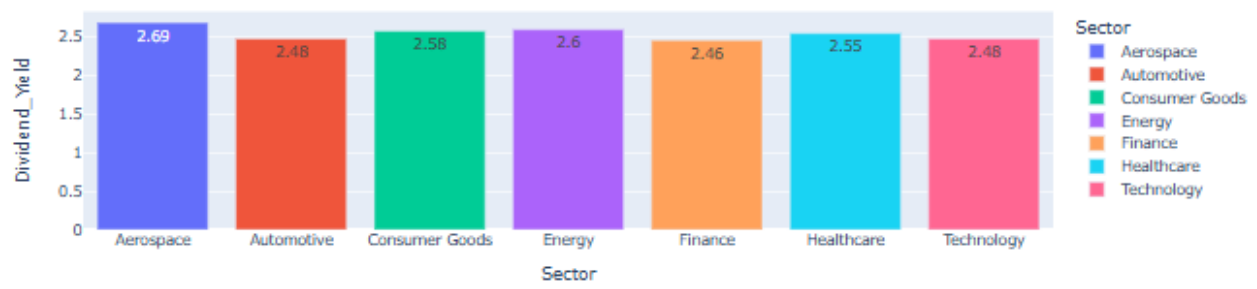
Average Close Price by Sector



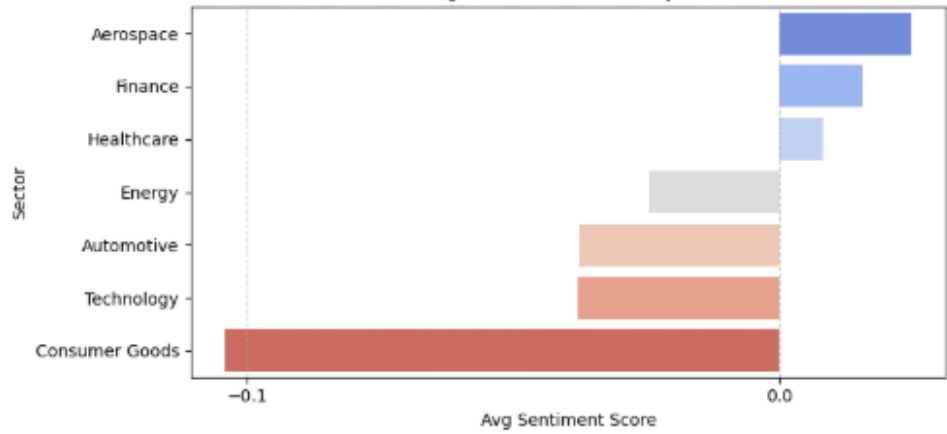
Average PE Ratio by Sector



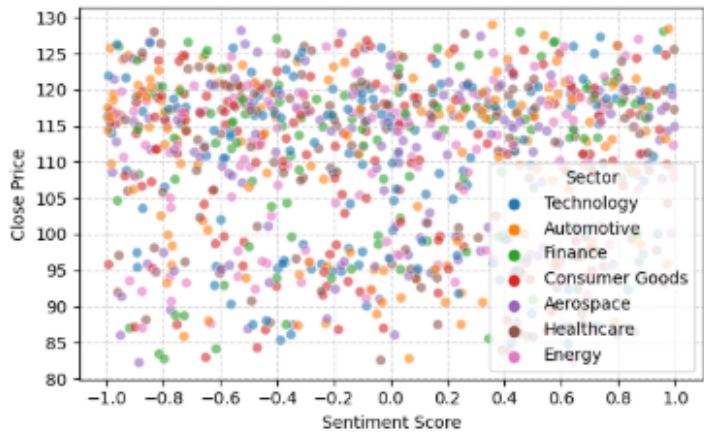
Average Dividend Yield by Sector



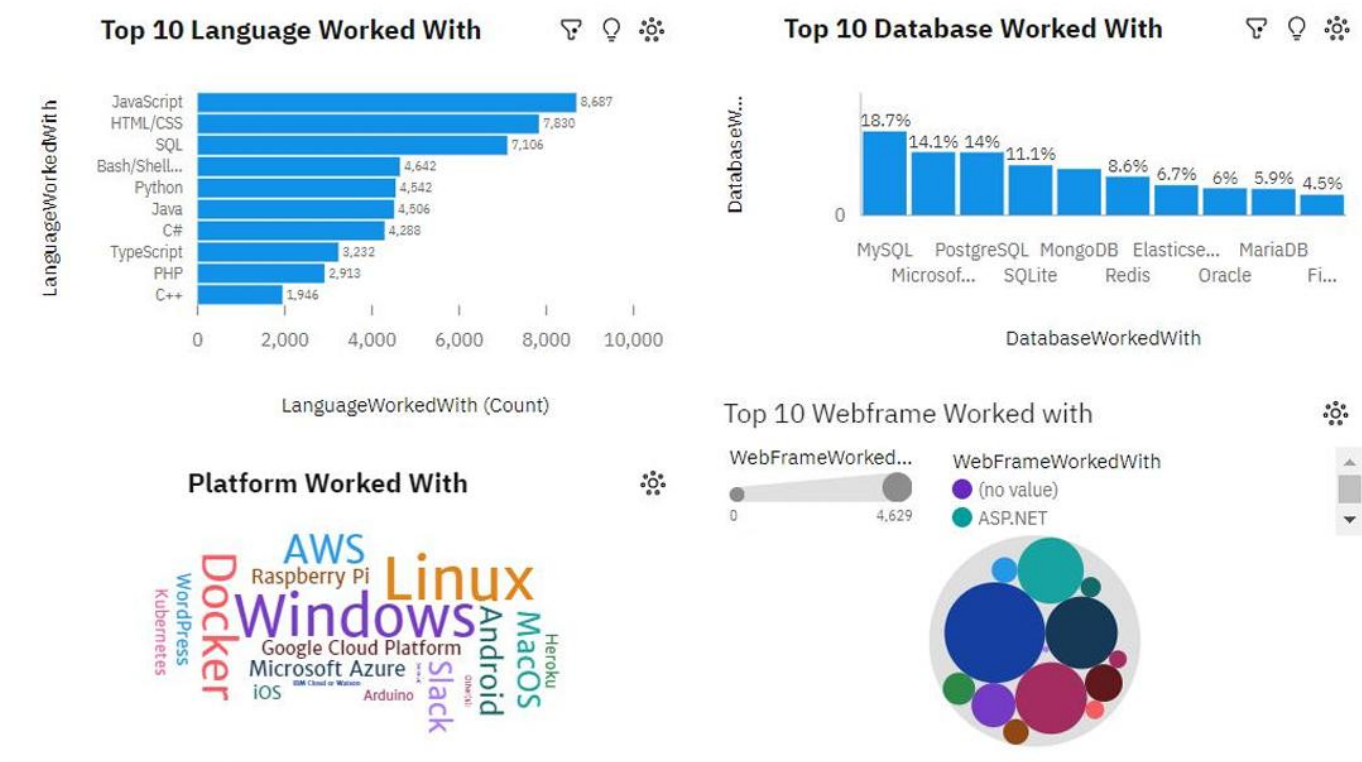
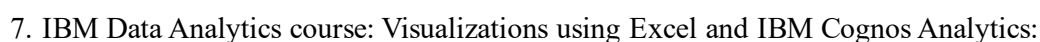
Average Sentiment Score by Sector

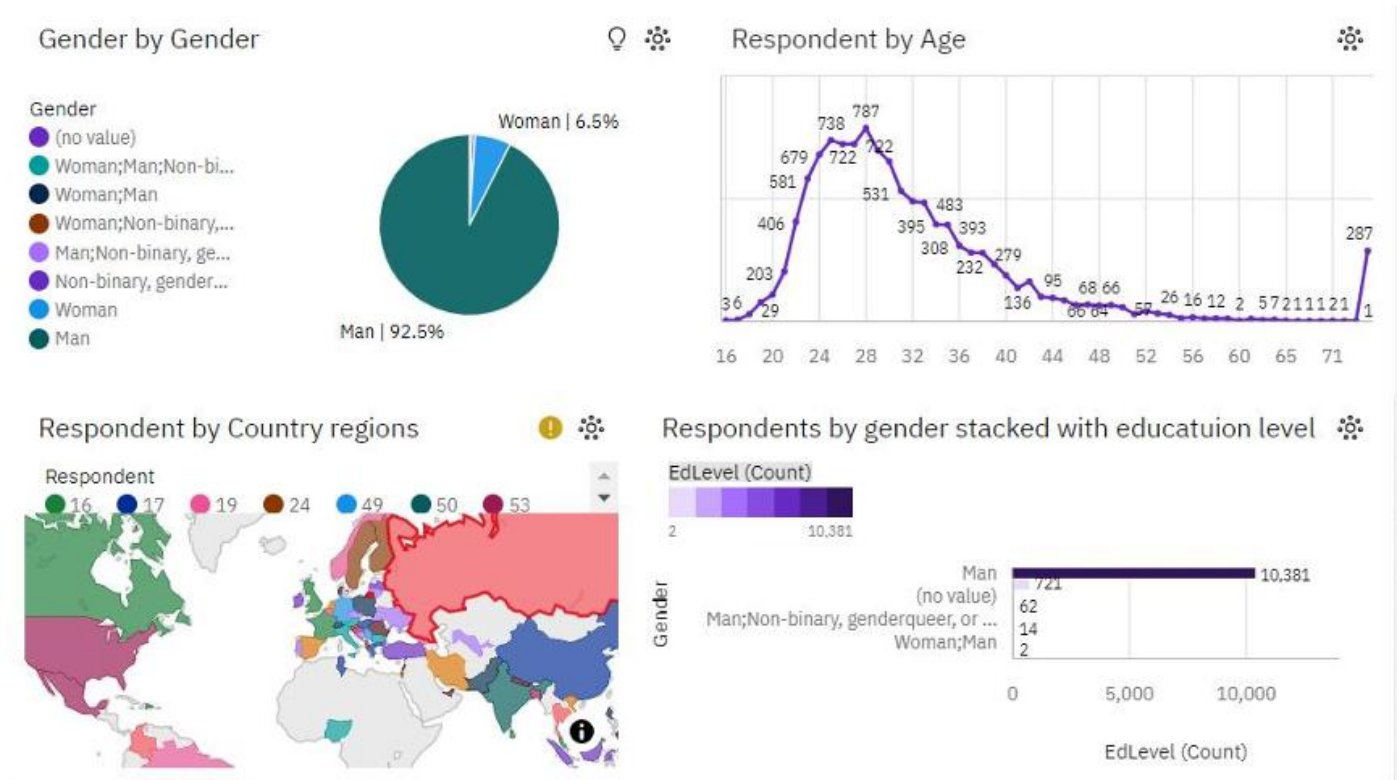


Sentiment Score vs Close Price



Tool used: MS Excel, MS Power BI



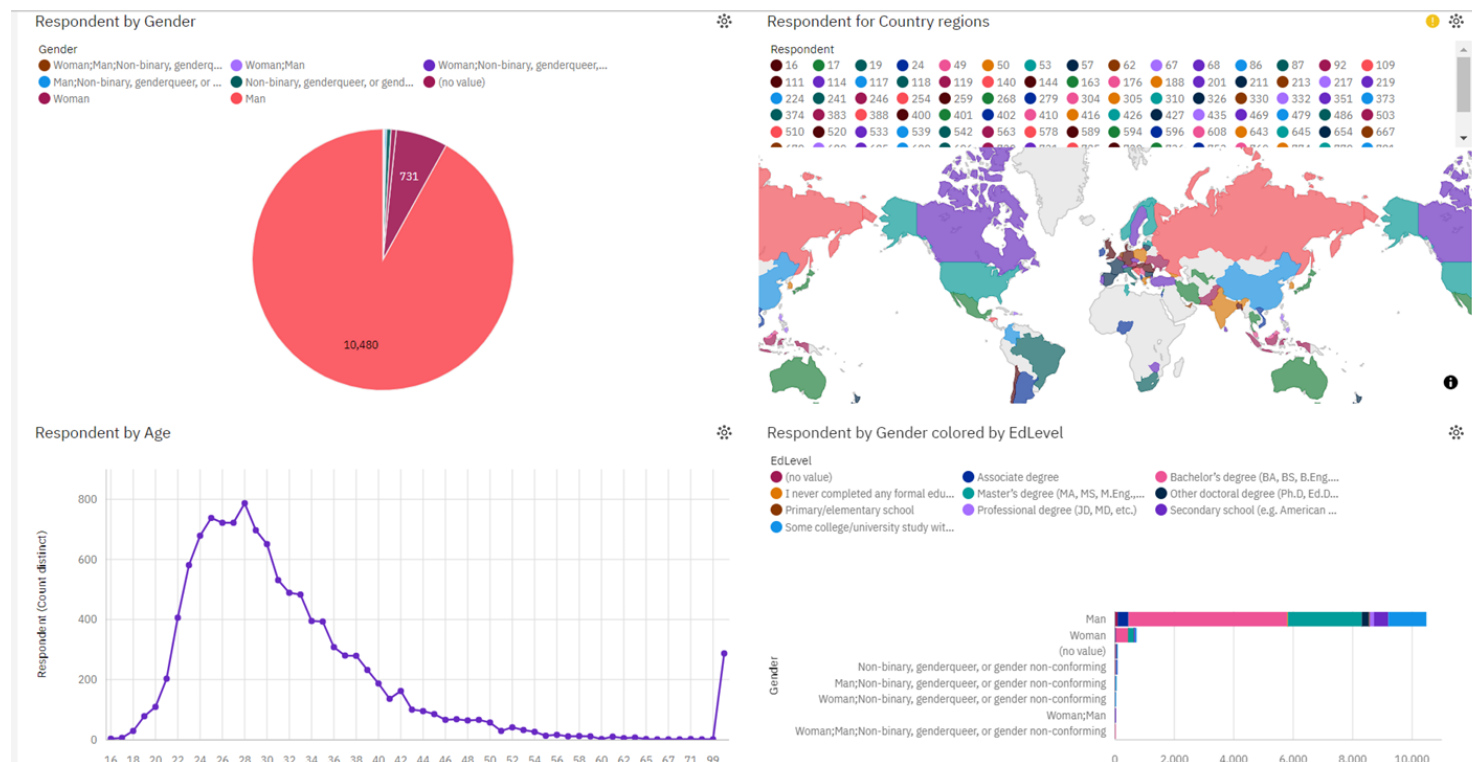


CERTIFICATION EARNED:



8. IBM Data Analytics course: CAPSTONE PROJECT.

DEMOGRAPHICS DASHBOARD:



CERTIFICATION EARNED:

